

Brandon Ng

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EDUCATION

B.S Data Science in Machine Learning and AI

University of California, San Diego

Expected Graduation: June 2026

GPA: 3.8

Relevant Coursework: Principles of Data Science, Statistical Methods, Practice of Data Science, Exploratory Data Analysis and Inference, Intro. Data Management, Discrete Mathematics, Vector Calculus, Linear Algebra, Data Structures and Algorithms, Computer Architecture, Probability and Statistics, Theoretical Foundations of Data Science I & II, Intro to Data Mining, Probabilistic Modeling and ML, Systems for Scalable Analytics, Representation Learning, Deep Learning for NLP

EXPERIENCES

MOSAIC Lab | Machine Learning Apprenticeship

Sept 2025 - ONGOING

- Engineering a multimodal cardiovascular digital twin dataset integrating 800K+ ECG recordings with EHR data (demographics, vital signs, lab results) from 160K+ patients, enabling patient-specific modeling of cardiac health trajectories
- Developed modular preprocessing pipeline performing static feature extraction, temporal alignment of clinical events to ECG timestamps, and cardiovascular entity extraction from 500K encounters across hospital admissions, ICU stays, and ED visits
- Designed temporal linkage framework matching ECG acquisitions to clinical encounters through time-windowed alignment, preserving chronological ordering for longitudinal cardiac monitoring and disease progression analysis

XR Lab | Machine Learning Research Assistant

Sept 2025 - ONGOING

- Led research initiative on VLMs and segmentation architectures for automated interior image tagging, evaluating Mask R-CNN, YOLO-World, SAM2, and GroundingDINO to identify optimal approaches for architectural attribute detection
- Implemented Stage 1 prototype using GroundingDINO achieving high-quality region proposals for furniture detection and spatial understanding, establishing foundation for scalable, cost-efficient tagging infrastructure, and limitations
- Researching depth estimation models (DPT/MiDaS) for dense per-pixel depth prediction to enable 3D spatial relationship analysis for room dimensions, ceiling heights, and architectural concavities in interior scenes

Comcast | Software Engineer Intern

June 2025 - Aug 2025

- Refactored and optimized a large-scale recommendation model, migrating from TensorFlow to PyTorch to enable a direct comparison of training efficiency, data pipelines, and library performance for scalable machine learning deployment
- Engineered preprocessing and distributed training pipelines, integrating Hugging Face and Petastorm with PyTorch DataLoaders, implementing a custom sampled softmax loss, and leveraging TorchDistributor for multi-node training
- Conducted A/B benchmarking of PyTorch vs. production TensorFlow models over a 7-day test, achieving higher relevance scores, faster training times, and improved content recommendations, while observing slightly narrower coverage

California Department of Public Health | Software Engineer Student Intern

Feb 2025 - May 2025

- Revamped R setup and RecordLinkage workflows, adding sample datasets and improved RStudio guidance, while researching RecordLinkage string comparison algorithms and weighting schemes to enhance duplicate detection and data quality
- Digitized and deployed the Lead Poisoning Follow-Up Form, transforming a legacy PDF into an interactive Power App, designing a structured SharePoint schema, and creating an end-to-end data pipeline to streamline data collection
- Spearheaded development of modernized backend workflows and data systems, analyzing RASCAL, migrating on-premise servers to the cloud via REDCap and Azure, and leveraging Power Automate to improve scalability and reduce manual effort

PROJECTS

Plant Disease Detector

Python, PyTorch, Pandas, CV

Developed a convolutional neural network using PyTorch to classify over 54,000 leaf images and detect 38 plant diseases across 14 species. Improved model performance by optimizing data pipelines and fine-tuning hyperparameters, achieving 89% accuracy rate

ChatGPT EDA

Python, Pandas, Scikit-learn

Performed exploratory data analysis to assess the impact of AI tools like ChatGPT on UCSD's academic environment. Used statistical tests and visualizations to uncover relationships between study habits, grades, and ChatGPT usage across departments

Culinary Insights

Python, Pandas, Scikit-learn

Analyzed recipes on Food.com and built a Random Forest model to predict recipe healthiness based on eight nutritional features. Enhanced model accuracy through grid search optimization and applied hypothesis testing to identify dietary trends within recipes

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, HTML, CSS, Java, C/C++, Swift, Assembly, R, SQL, OOP

Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch, Tensorflow, TorchDistributor, HuggingFace

Frameworks: Angular.js, React.js, Next.js, Node.js, Django, NoSQL, PostgreSQL

Tools: Git, Github, VSCode, Jupyter Notebook, Docker, Selenium, Postman, AWS S3, Apache Spark, Databricks